

STRING THEORY - JOSEPH POLCHINSKI

Reading Plan (2- quarters ^{of} courses worth):

Chapters 1-3, section 4.1, chapters 5-8 (omitting sections 5.4 & 6.7), chapters 10, sections 11.1, 11.2, 11.6, 12.1, and 12.2, and chapters 13 and 14.

Learning strategy: Writing notes & questions as I read and do, some of the exercises after each chapter.

Chapter 1, 1.1

Conceptual question: so 1-d objects give GR, consistent qGR, Grand Unification etc, but why can't this be done with 0-d objects i.e. points?
Also wouldn't fields technically be 4-d (3 spatial dim) objects even if the excitations happen in 0d i.e. on a point?

Note: While there is a unique theory (of strings), it has enormous numbers of classical solutions, even if we restrict attention to solutions with 4- large flat dimensions.

Q: What are weakly and strongly coupled string theories and how do they differ?

Q: Why is it said there is one unique string theory but there seems to be more based on the text?

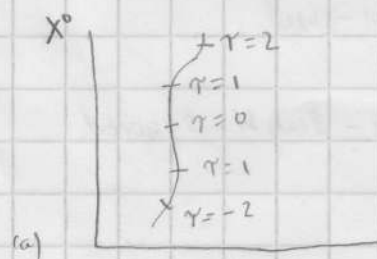
1.2 Action principle

String moves in D flat spacetime dimensions with ^{metric} $\eta_{\mu\nu} = \text{diag}(-, +, +, \dots, +)$.

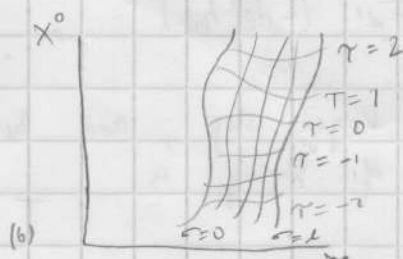
Point particle: motion by $D-1$ functions of time $X(X^0) \Rightarrow$ Hides covariance

$\Rightarrow$ Better: D functions of τ , a parameter along particle world line, $\Rightarrow X^\mu(\tau)$.

$\Rightarrow$ Arbitrary parameter i.e. $X'^\mu(\tau'(\tau)) = X^\mu(\tau) \Rightarrow X'^\mu$ & X^μ are same physically for any monotonic function $\tau'(\tau) \Rightarrow$ We get more symmetry but redundant description



a) point particle $\Rightarrow$ World line



b) Parametrized world sheet of an open string

The simplest Poincaré-invariant action which doesn't depend on the parametrization:

$$S_{pp} = -m \int d\tau (-\dot{X}^\mu \dot{X}_\mu)^{1/2}, \quad \text{where dot is a } \tau \text{ derivative}$$

Variation gives (after integration by parts): $\delta S_{pp} = -m \int d\tau \dot{u}_\mu \delta X^\mu$, where

$$u^\mu = \dot{X}^\mu (-\dot{X}^\nu \dot{X}_\nu)^{-1/2} \quad \text{is the normalized D-velocity,}$$

$\Rightarrow$ Eq of motion: $\dot{u}^\mu = 0$ describes free motion

Action can be put in another useful form by introducing additional field on the world line, an independent world line metric $\gamma_{\tau\tau}(\tau)$. Then we use the tetrad $\eta(\tau) = (-\gamma_{\tau\tau}(\tau))^{1/2}$ (is defined to be positive)

and get action as

$$S'_{pp} = \frac{1}{2} \int d\tau (\eta^{-1} \dot{X}^\mu \dot{X}_\mu - \eta m^2)$$

$\Rightarrow$ Same symmetry as S_{pp} : Poincaré & world-line reparametrization invariance

$$\Rightarrow \eta'(\tau') d\tau' = \eta(\tau) d\tau$$

EOM from varying η gives $\eta^2 = -\dot{X}^\mu \dot{X}_\mu / m^2$

$\Rightarrow S'_{pp}$ makes sense for massless particle, while S_{pp} doesn't work for that.

S_{pp} and S'_{pp} equivalent classically but S_{pp} hard to make sense in path integral

$\Rightarrow$ We use S'_{pp} as starting point for defining quantum theory. $\Rightarrow$ too complicated

One dimensional object:

- Sweeps 2d world sheet $\Rightarrow X^\mu(\tau, \sigma)$ as in figure 1b (two parameters σ, τ)

In relativistic quantum theory it is necessary to have parametrization independent

The Simplest action, Nambu-Goto action is proportional to the area of world sheet. For this we need the induced metric h_{ab} , $a \in \{\sigma, \tau\}$, defined by

$$h_{ab} = \partial_a X^\mu \partial_b X_\mu,$$

$\Rightarrow$ Nambu-Goto action: $S_{NG} = \int_M d\tau d\sigma L_{NG}$, with

$$L_{NG} = -\frac{1}{2\pi\alpha'} (-\det h_{ab})^{1/2}, \quad M \text{ denotes the world-sheet.}$$

α' is the Regge slope, and has units of spacetime-length-squared.

The tension of string is

$$T = \frac{1}{2\pi\alpha'}$$

The symmetries of the NG-action, the transformations of $X^\mu(\tau, \sigma)$ such that $S_{NG}[X'] = S_{NG}[X]$ ones

1. Isometry group of flat spacetime, the D-dimensional Poincaré group,

$$X'^\mu(\tau, \sigma) = \Lambda^\mu_\nu X^\nu(\tau, \sigma) + a^\mu, \text{ with } \Lambda^\mu_\nu \text{ a Lorentz transformation and } a^\mu \text{ a translation.}$$

2. Two-dimensional coordinate invariance, a diffeomorphism (diff) invariance. For new coordinates $(\tau'(\tau, \sigma), \sigma'(\tau, \sigma))$, the transformation is

$$X'^\mu(\tau', \sigma') = X^\mu(\tau, \sigma)$$

The NG-action is analogous to the point particle action S_{pp} , with derivatives in square root. $\Rightarrow$ We once again simplify it by introducing an independent world-sheet metric $\gamma_{ab}(\tau, \sigma)$.

Henceforth 'metric' will always mean γ_{ab} (unless we specify 'induced')

$\Rightarrow$ indices will be lowered or raised by this metric (always).

γ_{ab} has signature $(-, +)$. The action becomes:

$$S_p[X, \gamma] = -\frac{1}{4\pi\alpha'} \int_M d\tau d\sigma (-\gamma)^{1/2} \gamma^{ab} \partial_a X^\mu \partial_b X_\mu, \text{ where } \gamma = \det \gamma_{ab}.$$

This is Brink-Di Vecchia-Howe-Deser-Zumino action, or Polyakov action for short. The variation is

$$\delta S_p[X, \gamma] = -\frac{1}{4\pi\alpha'} \int_M d\tau d\sigma (-\gamma)^{1/2} \delta \gamma^{ab} (h_{ab} - \frac{1}{2} \gamma_{ab} \gamma^{cd} h_{cd}),$$

where we used $\delta \gamma = \gamma \gamma^{ab} \delta \gamma_{ab} = -\gamma \gamma_{ab} \delta \gamma^{ab}$.

$\Rightarrow \delta S_p = 0$ implies $h_{ab} = \frac{1}{2} \gamma_{ab} \gamma^{cd} h_{cd}$, dividing by $(-\gamma)^{1/2}$ gives $h_{ab} (-\gamma)^{-1/2} = \gamma_{ab} (-\gamma)^{-1/2}$ which can be used to eliminate γ_{ab} from action to recover $S_{NG}[X]$:

$$S_p[X, \gamma] \rightarrow -\frac{1}{2\pi\alpha'} \int d\tau d\sigma (-h)^{1/2} = S_{NG}[X]$$

The action S_p has the following symmetries:

1. D-dimensional Poincaré invariance: $X'^\mu(\tau, \sigma) = \Lambda^\mu_\nu X^\nu(\tau, \sigma) + a^\mu$,
 $\gamma'_{ab}(\tau, \sigma) = \gamma_{ab}(\tau, \sigma)$

2. Diff invariance: $X'^\mu(\tau', \sigma') = X^\mu(\tau, \sigma)$, $\frac{\partial \sigma'^c}{\partial \sigma^a} \frac{\partial \sigma'^d}{\partial \sigma^b} \gamma'_{cd}(\tau', \sigma') = \gamma_{ab}(\tau, \sigma)$
 for new coordinates $\sigma'^a(\tau, \sigma)$.

3. Two dimensional Weyl invariance: $X'^\mu(\tau, \sigma) = X^\mu(\tau, \sigma)$,

$$\gamma'_{ab}(\tau, \sigma) = \exp(2w(\tau, \sigma)) \gamma_{ab}(\tau, \sigma), \quad \text{for arbitrary } w(\tau, \sigma).$$

The Weyl invariance, a local rescaling of world-sheet metric, has no analog in the Nambu-Goto form. This is because $h_{ab}(-h)^{-1/2} = \gamma_{ab}(-\gamma)^{-1/2}$ doesn't completely determine γ_{ab} but only up to a local rescaling, so Weyl-equivalent metrics correspond to the same embedding in spacetime. $\Rightarrow$ Extra redundancy of Polyakov formulation, in addition to the obvious diff invariance.

The variation of the action with respect to the metric defines the energy-momentum tensor:

$$T^{ab}(\tau, \sigma) = -4\pi (-\gamma)^{-1/2} \frac{\delta}{\delta \gamma_{ab}} S_P = -\frac{1}{\alpha'} (\partial^a X^\mu \partial^b X_\mu - \frac{1}{2} \gamma^{ab} \partial_c X^\mu \partial^c X_\mu).$$

(has extra factor of -2π vs a usual def. in field theory; standard in string theory)

$$\nabla_a T^{ab} = 0 \quad \Rightarrow \text{Conserved as a consequence of diff invariance.}$$

Invariance of S_P under Weyl transformations implies $\gamma_{ab} \frac{\delta}{\delta \gamma_{ab}} S_P = 0 \Rightarrow T^a_a = 0$

So S_{NG} & S_P define 2D field theories on the ^{string} world-sheet.

In string theory amplitudes for spacetime processes are given by matrix elements in 2D qft on the world-sheet.

While interest is 4D spacetime, most machinery ^{we use} of string perturbation theory is 2D. From world-sheet P.O.V. coord transform $X'^\mu(\tau', \sigma') = X^\mu(\tau, \sigma)$ defines $X^\mu(\tau, \sigma)$ as a scalar field, with μ as an internal index.

Q: What does internal index mean and how it differs from external?

$\Rightarrow$ From 2D view Polyakov action describes massless Klein-Gordon scalars X^μ covariantly coupled to γ_{ab} . From its POV, the Poincaré invariance is an internal symmetry $\Rightarrow$ it acts on fields at fixed τ, σ .

Q: What is internal vs external symmetry and how do they differ (mathematically or physically)?

Varying γ_{ab} in the action gives the EOM: $T_{ab} = 0$.

Varying X^μ gives the EOM: $\partial_a [(-\gamma)^{1/2} \gamma^{ab} \partial_b X^\mu] = (-\gamma)^{1/2} \nabla^2 X^\mu = 0$.

World sheet with boundary $\rightarrow$ surface term in variation of action.

Take the coordinate region to be

$$-\infty < \tau < \infty, \quad 0 \leq \sigma \leq \ell. \quad \left(\text{think of } \tau \text{ as time variable and } \sigma \text{ as spatial} \right)$$

($\Rightarrow$ single string propagating without sources.)

$$\rightarrow S_P = \frac{1}{2\pi\alpha'} \int_{-\infty}^{\infty} d\tau \int_0^{\ell} d\sigma (-\gamma)^{1/2} S X^\mu \nabla^2 X_\mu$$

$$- \frac{1}{2\pi\alpha'} \int_{-\infty}^{\infty} d\tau (-\gamma)^{1/2} S X^\mu \partial^\sigma X_\mu \Big|_{\sigma=0}^{\sigma=\ell}$$

The boundary vanishes if

$$\partial^\sigma X^\mu(\tau, 0) = \partial^\sigma X^\mu(\tau, \ell) = 0$$

$\Rightarrow$ Neumann boundary condition, can also be written as (more covariantly):
 $n^a \partial_a X_\mu = 0$ on ∂M , n^a is normal to ∂M . The ends of string move freely in spacetime

The surface term also vanishes for periodic boundary condition (strings form a closed loop):
 $X^\mu(\tau, \ell) = X^\mu(\tau, 0)$, $\partial^\sigma X^\mu(\tau, \ell) = \partial^\sigma X^\mu(\tau, 0)$, $\gamma_{ab}(\tau, \ell) = \gamma_{ab}(\tau, 0)$.

The open string boundary condition (Neumann) and the closed string boundary condition are the only possibilities consistent with D -dimensional Poincaré invariance.

In QFT Symmetry, and not simplicity, is the key idea $\Rightarrow$ We must consider most general local action consistent with all of the symmetries of the theory.

Q: Why we started from simplicity (i.e. simplest action) then yank their symmetry and now construct most general action with said symmetry instead of first figuring out the symmetries, since this seems quite arbitrary?

$\Rightarrow$ We now try generalise the Polyakov action (all symmetries still maintained) with polynomial action (no polynomial $\rightarrow$ hard to make sense of qft)

Global Weyl invariance: Action must have one more factor of γ^{ab} than γ_{ab} to cancel the variation of $(-\gamma)^{1/2}$
 This + coordinate invariance and Poincaré invariance gives us one additional term beyond S_P :

$$X = \frac{1}{4\pi} \int_M d\tau d\sigma (-\gamma)^{1/2} R$$

where R is Ricci scalar constructed from γ_{ab} .

Under a local Weyl rescaling, we have

$(-\gamma')^{1/2} R' = (-\gamma)^{1/2} (R - 2\nabla^2 w)$, The variation is a total derivative,
 because $(-\gamma)^{1/2} \nabla_a v^a = \partial_a ((-\gamma)^{1/2} v^a)$ for any v^a . The integral above is therefore invariant for a world sheet without boundary. With boundaries on

additional surface term is needed.

Since χ is allowed by the symmetries we will include it in the action:

$$S'_P = S_P - \lambda \chi$$
$$= - \int_M d\tau d\sigma (-\gamma)^{1/2} \left(\frac{1}{4\pi\alpha'} \gamma^{ab} \partial_a X^\mu \partial_b X_\mu + \frac{\lambda}{4\pi} R \right)$$

This is the most general (Diff $\times$ Weyl) - invariant and Poincaré invariant action with these fields and symmetries.

$\Rightarrow$ Note: so far the symmetries are classical and we are ignoring quantum anomalies.

$\Rightarrow$ S'_P looks like the Hilbert action for the metric, $\int (-\gamma)^{1/2} R$, minimally coupled to D massless scalar fields X^μ .

$\Rightarrow$ In 2D Hilbert action doesn't give dynamics, it only depends on the topology of world sheet

There are also other ways to generalize the theory:

- interesting case for us is $D=4$, so allow more than 4 fields X^μ but require only 4 have Poincaré invariance.
- Second idea is to keep same local symmetries & Poincaré invariance, but with more general field content $\Rightarrow$ 2D fields in different representations of the world-sheet and spacetime symmetry.
- A further idea is to enlarge the local symmetry of theory

1.3 The open string spectrum

This section has emphasis on results, and not the methods, so should not dwell in fine points since systematic treatment in next chapter

$\Rightarrow$ No notes for this section, just plain reading.